

Name: _____ Student NUID: _____

CS 2100: Program Design and Implementation 1

Practice Quiz 2

Instructions

- Do not begin the quiz until instructed to do so.
- You may use both sides of a sheet of paper up to 8.5"x11" for reference, but no other resources, including phones, computers, AI, headphones, and ear pods.
- You have until the end of the class period to complete the quiz.
- Students may not leave the classroom during the first 10 minutes of the quiz (except in case of emergency).
- Hand your completed quiz to an instructor before leaving the room.
- Talk to an instructor if you need to leave the room and reenter.

1. Git (revisited from Quiz 1)

Consider the scenario where you run `git add homework3.py`, and then make a few more edits to `homework3.py` before committing. When you run `git status`, you see `homework3.py` listed under both `Changes to be committed` and `Changes not staged for commit`. Why is it listed in both sections?

Solution:

The changes done before the first `git add` command have been staged for commit, and they are included in the version listed under `Changes to be committed`. The changes after that `git add` are not yet staged for commit.

If a file is listed under `Changes not staged for commit`, then that means it has new changes that are not staged for commit, and those changes need to be `git add`ed to be staged.

2. Functions and Documentation (revisited from Quiz 1)

Please write appropriate and accurate documentation for this function:

```
def multiply_substring(source: str, substring: str, multiplier: int) -> str:
    """
    """

    if multiplier < 0:
        raise ValueError("Multiplier cannot be negative")
    result = ""
    i = 0
    while i < len(source):
        end_index = i + len(substring)
        window: str = source[i : end_index] # substring starting at index i, ending at end_index
        if window == substring:
            result += substring * multiplier
            i += len(substring)
        else:
            result += source[i]
            i += 1
    return result
```

Solution:

Returns the source string, with all substrings that match the substring multiplied by the multiplier.

Parameters:

source : str
The string within which to multiply substrings
substring : str
The substring to search and multiply
multiplier : int
The number of times to multiply each substring

Returns:

str
The modified string with multiplied substrings

Raises:

ValueError
If the multiplier is negative

3. Classes

Read the `ProtectedPerson` class below.

Please add appropriate methods to the `ProtectedPerson` class so that if `person` is a `ProtectedPerson`, then

`print(person)` prints:

- `<name>` is unprotected if their bodyguard is `None`, and
- `<bodyguard-name>` protects `<name>`, where `<bodyguard-name>` is replaced by their bodyguard's name, if they have a non-`None` bodyguard

where `<name>` is `person`'s name.

```
class ProtectedPerson:
    """A class representing a person with a bodyguard"""
    def __init__(self, name: str, bodyguard: Optional[str]) -> None:
        self.name = name
        self.bodyguard = bodyguard
```

Solution:

```
def __str__(self) -> str:
    if self.bodyguard is None:
        return f"{self.name} is unprotected"
    else:
        return f"{self.bodyguard} protects {self.name}"
```

4. Aliasing and Equality

Using the `ProtectedPerson` class from the previous question, what is the output of this code? Assume there is no other code in the `ProtectedPerson` class outside of the previous question.

```
blob = ProtectedPerson("Blob", "Glob")
glob: ProtectedPerson = blob
print(blob == glob)
```

Solution:

True

5. Stakeholder-value Matrices

Consider an online shopping website where people can order clothing online to be shipped to them. Please fill in the cells in the bottom row of the following Stakeholder-value matrix, where the stakeholder is the shipping worker.

	Shipping distance	Physical safety
Customer	May want to reduce distance to minimize carbon footprint	May want to prevent their shipping address from being made available to others
Shipping worker		

Solution:

Shipping distance: Likely wants to reduce time spent travelling

Physical safety: Likely does not want to delivery to dangerous locations